Exercise 1

Exercise 1 — Revenues vs collections, expenses vs payments, and depreciation

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A company started activity on 1 January of the present year. At 31 March it has the following goods, acquired on 31 December of the previous year:

- *factory building 1,000 m.u., duration 25 years;*
- *office building 1,500 m.u., duration 50 years;*
- *machinery 300 m.u., duration 10 years;*
- *transport equipment 100 m.u., duration 5 years.*

Information for the first quarter:

- *personnel salaries of the quarter 450 m.u., of which 75 m.u. are pending to be paid after 31/3;*
- *materials purchased and used in production 700 m.u., of which 125 m.u. are pending to be paid after 31/3;*
- *energy consumption estimated at 230 m.u., nothing paid to date;*
- *sales invoiced to customers 1,930 m.u., of which 315 m.u. had not been collected at 31/3; the rest was collected in due time. In manufacturing all the units sold, all the purchased materials were consumed.*

Ignore corporation taxes.

a) At 31/3, find the income, expenses, collections (cash inflows), payments (cash outflows), investments and depreciation of the quarter (linear depreciation along the lifetime of the goods).

b) How does the answer change if only half of the purchased materials had been consumed in manufacturing the units sold?

Phase 1 — Setup: data and hypotheses

Period: first quarter (1/01 to 31/03), $\frac{3}{12} = \frac{1}{4}$ of a year. The fixed assets were bought on 31/12 of the previous year, so they are used during the **whole** quarter.

The exercise trains the distinction of [Slide 8]: a **revenue** is recorded when the sale is made, whether or not it is collected; an **expense** when the good or service is received or consumed, whether or not it is paid. Collections and payments are the cash side of the same operations.

Phase 2 — Frames: classification of every operation

Operation	Revenue / expense of Q1	Collection / payment in Q1	Pending at 31/3
Sales	Revenue 1,930	Collection $1{,}930 - 315 = 1{,}615$	Receivable 315
Salaries	Expense 450	Payment $450 - 75 = 375$	Salaries payable 75
Materials (all consumed)	Expense 700	Payment $700 - 125 = 575$	Supplier payable 125
Energy	Expense 230	Payment 0	Energy payable 230
Fixed assets	Depreciation 30	0	Net value $2{,}900 - 30$

Phase 3 — Derivation

Step 1. Depreciation of the quarter

For each asset, with no residual value given:

$$D_{\text{quarter}} = \frac{\text{Cost}}{\text{Years of life}} \times \frac{3}{12}$$

Asset	Annual depreciation	Quarter
Factory building	$1{,}000/25 = 40$	$40/4 = 10.0$
Office building	$1{,}500/50 = 30$	$30/4 = 7.5$
Machinery	$300/10 = 30$	$30/4 = 7.5$
Transport equipment	$100/5 = 20$	$20/4 = 5.0$
Total	**120**	**30.0**

Step 2. Part (a)

- **Income (revenues):** the sales invoiced, **1,930 m.u.**
- **Expenses:** $450 + 700 + 230 + 30 =$ **1,410 m.u.** (so the profit of the quarter is $1{,}930 - 1{,}410 = 520$ m.u.).
- **Collections:** $1{,}930 - 315 =$ **1,615 m.u.**
- **Payments:** $(450 - 75) + (700 - 125) + 0 = 375 + 575 =$ **950 m.u.** Depreciation is not a payment.
- **Investments of the quarter: 0 m.u.** The fixed assets (total $1{,}000 + 1{,}500 + 300 + 100 = 2{,}900$ m.u.) were acquired on 31/12 of the previous year, before the quarter began. If the question is read as "the investment the company works with", the answer is that 2,900 m.u.; state the reading you use in the exam.
- **Depreciation: 30 m.u.**

Step 3. Part (b): only half of the materials consumed

Only the material **consumed** is an expense [Slide 8, Slide 10]: $700/2 = 350$ m.u. The other 350 m.u. stay in the warehouse as **inventory**, a current asset [Slide 14].

Magnitude	(a)	(b)	Change
Income	1,930	1,930	0
Expenses	1,410	$450 + 350 + 230 + 30 = 1{,}060$	–350
Profit	520	870	+350
Collections	1,615	1,615	0
Payments	950	950	0
Inventory at 31/3	0	350	+350
Depreciation	30	30	0

Payments do not change: the company **paid** for what it **bought** (575 of the 700), regardless of how much it **used**.

Phase 4 — Interpretation and checks

- **Profit is not cash:** in (a) the profit is 520 while the net cash generated is $1{,}615 - 950 = 665$. The difference, $665 - 520 = 145$, is explained item by item: $+30$ (depreciation, an expense without payment) $+75 + 125 + 230$ (expenses not yet paid) -315 (revenue not yet collected) $= 145$.
- **Part (b)** raises the profit by 350 without any extra cash: unconsumed material is an asset, not an expense.
- **Check of depreciation:** the annual total 120 times $\frac{1}{4}$ gives 30.